

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 3

For the tutorial in Week 4 on 8 February, let us discuss

- Ex. 6.4.2, 6.4.5, 6.4.13, 6.4.16, 6.4.20, 6.4.21, Additional Exercises, 6.5.7, 6.5.9.

Ex. 6.4.2 and Ex. 6.4.5.

- (a) Find the coefficient of x^5y^8 in $(x+y)^{13}$.
(b) Give a formula for the coefficient of x^k in the expansion of $(x + \frac{1}{x})^{100}$, where k is an integer.

Solution:

- (a) $C(13, 8) = C(13, 5)$, by binomial theorem.
(b) For $0 \leq i \leq 100$, the i -th term of the expansion is

$$C(100, i) \cdot x^i \cdot \frac{1}{x^{(100-i)}} = C(100, i) \cdot x^{2i-100}.$$

Hence, when $2i - 100 = k$, we have $i = (100 + k)/2$, and the coefficient of x^k is

$$\begin{cases} C(100, \frac{100+k}{2}), & \text{for } -100 \leq k \leq 100, k \text{ even;} \\ 0, & \text{otherwise.} \end{cases}$$

Ex. 6.4.13. Prove that if n and k are integers with $1 \leq k \leq n$, then $k \binom{n}{k} = n \binom{n-1}{k-1}$

- (a) using a combinatorial proof. [Hint: Show that the two sides of the identity count the number of ways to select a subset with k elements from a set with n elements and then an element of this subset.]
(b) using an algebraic proof based on the formula for $\binom{n}{r}$.

Solution:

- (a) Both sides of the identity count the number of ways to select a subset with k elements from a set with n elements and then a distinguished element of this subset.

For the left side, we first select k elements out of the n elements to form the subset (this can be done in $\binom{n}{k}$ ways), and from the k elements, we choose a distinguished element (this can be done in k ways).

For the right side, we first choose the distinguished element from the n elements (this can be done in n ways), and then we select the remaining $k - 1$ elements of the subset from the remaining $n - 1$ elements (this can be done in $\binom{n-1}{k-1}$ ways).

- (b)

$$k \binom{n}{k} = k \frac{n!}{k!(n-k)!} = k \frac{n \cdot (n-1)!}{k \cdot (k-1)!(n-k)!} = n \frac{(n-1)!}{(k-1)!(n-k)!} = n \binom{n-1}{k-1}.$$

Ex. 6.4.16. Show that if p is a prime and k is an integer such that $1 \leq k \leq p-1$, then p divides $\binom{p}{k}$.

Solution: Note that

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

Clearly p is a factor of the numerator. On the other hand, each of these numbers $1, 2, \dots, k$, and $1, 2, \dots, p-k$ is less than p and so the prime factorization of the denominator only contains numbers less than p . As a consequence, p does not divide the denominator. Since $\binom{p}{k}$ is an integer, its prime factorization contains p and so p divides $\binom{p}{k}$.

Ex. 6.4.20. Show that if n is a positive integer, then $C(2n, 2) = 2C(n, 2) + n^2$

- (a) by algebraic manipulation;
- (b) using a combinatorial argument.

Solution:

- (a)

$$\begin{aligned} 2C(n, 2) + n^2 &= 2 \frac{n(n-1)}{2} + n^2 \\ &= \frac{2n(2n-1)}{2} = C(2n, 2). \end{aligned}$$

- (b) Consider a set A of n boys and another set B of n girls, we want to choose 2 people to complete a job — there are $C(2n, 2)$ different choices. On the other hand, we can choose the two people from A — $C(n, 2)$ choices, or from B — $C(n, 2)$ choices, or one from A and one from B — $C(n, 1) \cdot C(n, 1) = n^2$ choices. Hence, in total there are $2C(n, 2) + n^2$ choices. Therefore, $C(2n, 2) = 2C(n, 2) + n^2$.

Ex. 6.4.21. Give a combinatorial proof that $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$.

[Hint: Count in two ways the number of ways to select a committee and to then select a leader of the committee.]

Solution: Here are two ways of counting the committee:

- Select a committee of size $k \geq 1$, there are $\binom{n}{k}$ ways. Then out of the k members, select one as the leader (there are k ways), hence overall there are $k \cdot \binom{n}{k}$ ways. Summing over all possible k , there are $\sum_{k=1}^n k \binom{n}{k}$ ways.
- Select the leader first, followed by the other members of the committee. There are n ways to select the leader, and then out of the remaining $n-1$, select a subset of any size as the rest members of the committee, i.e., 2^{n-1} ways, hence in total $n \cdot 2^{n-1}$ ways.

Additional Exercise 1. How many seating arrangements are there with six boys and six girls if the table is triangular with 4 seats on each side? Two arrangements are identical if they differ by a turn of the table that keeps the chairs in place.

Solution:

There are $12!$ permutations of the 12 people. Mark a designated chair on the table. Consider the function that takes a permutation of the 12 people into a table arrangement by seating the first person on the designated chair and arranging the rest clockwise. This is a 3-to-1 function from the set of permutations to the set of seating arrangements, as each seating arrangement arises from the three permutations that differ by a 120 degree turn of the designated chair. Therefore there are $12!/3 = 159,667,200$ possible seating arrangements.

Additional Exercise 2. How many seating arrangements are there with six boys and six girls if the table is hexagonal with 2 seats on each side, out of which one must be occupied by a boy and the other one by a girl?

Solution:

Let us first count the number of such seating arrangements in which one table position is distinguished (say that chair is red while the other chairs are blue). We count the number of arrangements using the generalized product rule. For each of the 6 pairs of people that sit on the same side there are two possible gender assignments. Once a gender is assigned to each seat, there are $6!$ ways to assign the boys and $6!$ ways to assign the girls, so the total number of seating arrangements with one distinguished table position is $2^6 \cdot (6!)^2$. The function that erases the distinguished position is a 6-to-1 map, so the number of circular arrangement in which every boy sits across from a girl is $2^6 \cdot (6!)^2 / 6 = 5,529,600$.

Ex. 6.5.7, and Ex. 6.5.9.

- (a) How many ways are there to choose eight coins from a piggy bank containing 100 identical pennies and 80 identical nickels ?
- (b) A book publisher has 3000 copies of a discrete mathematics book. How many ways are there to store these books in their three warehouse if the copies of the book are indistinguishable, and each warehouse should store at least one book ?

Solution:

- (a) There are 9 ways, i.e., i pennies and $8 - i$ nickels for $i = 0, 1, \dots, 8$.
- (b) Let the number of books at each warehouse be x_1, x_2, x_3 , where they are positive integers and $x_1 + x_2 + x_3 = 3000$. Let $y_i = x_i - 1$ for $i = 1, 2, 3$, then $y_1 + y_2 + y_3 = 3000 - 3 = 2997$. There are $C(2997 + 3 - 1, 2997) = C(2999, 2) = 4,495,501$ solutions.